Which new study programme should OTH offer?

A plain-English summary of the Study Program Intelligence project

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1. The answer on one page

OTH Amberg-Weiden should start a bachelor's degree in cybersecurity in its faculty of Electrical Engineering, Media and Computer Science (EMI). We compared three possible new programmes using only public data. The cybersecurity bachelor came first in 91.7 % of 10,000 what-if scenarios. Our middle estimate is about 39 first-year students a year once the programme is fully running. In our model about four in five of them would be new to OTH. The biggest uncertainty is how many students similar programmes really attract, because none of the similar programmes at universities of applied sciences that we checked publishes this number.

Question	What we found	Key number
Do students want it?	Computer science is the faculty's strongest large subject, but most of its growth in Bavaria comes from foreign students.	6.1 % a year in Bavaria; German students only 2.3 % a year
Would they come to OTH?	OTH has its own region; the nearest full-time cyber bachelors are 82 km and 105 km away.	21 / 39 / 75 a year once running (first year 12 / 24 / 45)
Would they be new to OTH?	In our model most would otherwise study elsewhere.	82 % in the middle case
Can OTH teach it?	Much of the basics, little of the security core.	up to 71 % of the basics, about 31 % of security
Are there jobs?	Many adverts mention "Cyber Security", also in the wider region.	851 in Germany, 191 within 100 km of Amberg
How sure are we?	First place holds, by less if we judge demand on German students only; student numbers are uncertain.	first in 91.7 % of the scenarios (78.2 % judging demand on German students)

What OTH would need (our estimate): 2 new professorships, a lab engineer, outside teachers for law and security management, and a separate security lab whose cost we have not estimated. **What to do first:** ask TH Ingolstadt and TH Deggendorf for their real first-year numbers. One measured number would make our student estimate much more precise.

2. The question and why it matters

Our course gave us a clear task. Using only public data, we had to find which new study programme would attract the most students to OTH, and explain why it suits our faculty.

This matters because, across Germany, most of the faculty's electrical and media subjects attract fewer new students each year. Between 2019 and 2025 the number of first-year students changed by -2.6 % a year in electrical engineering and by -8.6 % a year in media technology. (2025 was a low year in Bavaria because of a school reform.) Computer science is the exception. It grew over this period, although in Germany it has stayed about level since 2023.

A new programme is a big decision. It needs new professors, rooms and equipment, and it takes years to show results. So the faculty needs more than a good feeling. It needs to know whether enough students would come, whether they would be new to OTH, whether the faculty can teach the subject, and whether graduates would find jobs.

We looked at bachelor's and master's programmes and at all of Germany for demand and competition. We used no internal OTH data. All sources are public, and our project files record where each number comes from and when we took it.

3. The big picture: four questions

Think of a company that wants to open a new shop. It would ask four things before it signs a lease. Do people want the product? Would they come to this shop, or go to a rival? Can the company make the product with its staff? And is there a future for the business? We asked the same four questions about a new programme:

1. **Do students want it?** We looked at how many people start studying related subjects each year, and how this may change until 2035.
2. **Would they come to OTH?** We looked at how many school leavers finish school near OTH, where the nearest similar programmes are, how far students usually travel, and how many students similar programmes plan for.
3. **Can OTH teach it?** We compared the course plans of similar programmes with what OTH's faculty already teaches. We estimated the extra staff and equipment, but not the cost of the lab.
4. **Are there jobs?** We counted job adverts and looked at employment statistics.

We answered these questions in full for the cybersecurity bachelor and with less evidence for the other two finalists. Earlier candidates were screened on student demand and competition, and two of them also on the overlap rule.

Two more checks came after the four questions. First, we asked whether a new programme would only take students away from OTH's existing programmes. We call this **cannibalisation**. It is like a new café that mainly takes customers from the owner's other café down the street. Second, we compared the options in **what-if scenarios**. We repeated the comparison 10,000 times. Each time, the weights of the six scoring points changed a little, and each score was drawn from its range, from low to high. Then we counted how often each option came first.

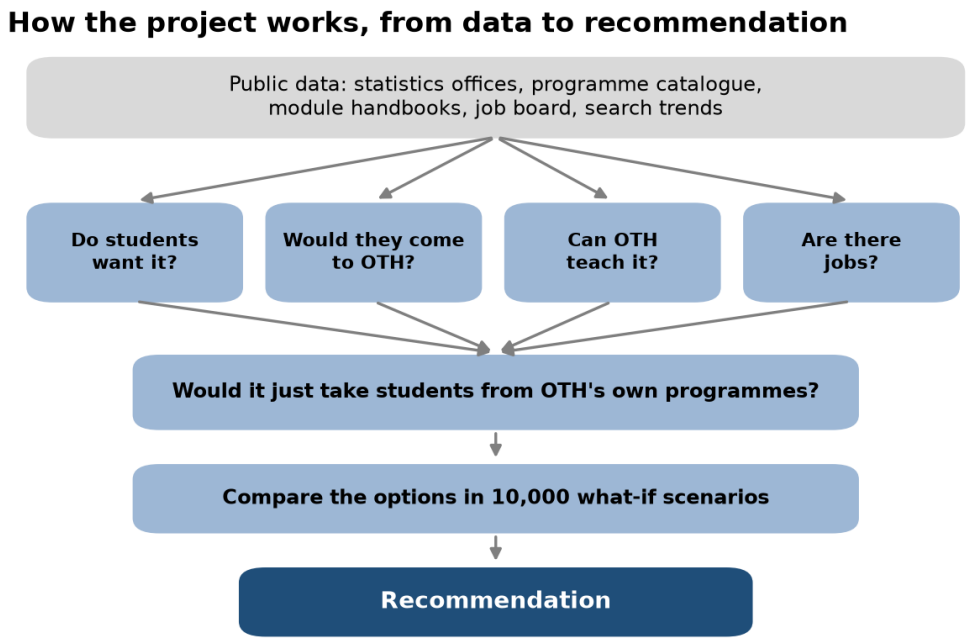


Figure 1: The whole project in one picture: public data answers four questions, then we check for lost students and compare the options.

Before the first screening, we fixed how much each of the six scoring points would count. We also fixed five **kill rules**. A kill rule drops an option at once. One example: more than half of its core topics are already taught in one existing OTH programme at the same level. Fixing these rules early makes it harder for our own preferences to shape the result. Some rules came later, and Section 5 says which.

4. Where the data came from

We used only public sources. Most of them come from public offices. Our core counts are official statistics, which count everyone. A few sources are weaker signals, and we use them only for comparison or for our ranges.

Source	What it tells us	Why we trust it
Federal Statistical Office (Destatis)	how many people start studying each subject, and how many graduate, 2005-2025	official count of all students starting in the winter semester

Source	What it tells us	Why we trust it
Bavarian State Office for Statistics	school leavers per county; where the students of each Bavarian university of applied sciences come from; IT jobs in Bavaria	official counts
Forecasts of the education ministers (KMK)	how many school leavers and first-year students to expect until 2035 and beyond	official forecasts of the education ministers
Study search of the Federal Employment Agency	all 4,840 bachelor and master offers in 13 study fields	national catalogue; we checked our keyword search by hand (94 % correct for cybersecurity)
Module handbooks	what each programme teaches, module by module	the universities' own official course plans
Job board and shortage report of the Federal Employment Agency	job adverts by keyword and region; jobs with too few applicants	official job board; the counts hardly changed between two days
Google Trends	how often people search for a degree subject	rough signal only, used to compare subjects
One student review	about ~60 first-year students in cybersecurity at TH Deggendorf	weak; it sets our high estimate, and so also our middle one

Three rules applied to all data. First, we respected each website's rules for automatic downloads. Second, where a site blocked downloads or needed a login, we did not work around it. We used our own accounts where we had access, downloaded public files by hand where only automatic downloads were blocked, or left the data out. Third, every statistic in this summary comes from one generated list of **key numbers**, which names the document each number comes from.

5. How we narrowed down the options

Imagine a hiring panel that writes down its rules before it reads any application. Then nobody can say the panel picked its favourite first and invented the rules afterwards. We did the same for the weights and the kill rules. On 14 Oct 2026 we first fixed the six scoring points with their weights, then the five kill rules. The project history shows that the kill rules came about five hours before we screened the first data on study fields.

Then we narrowed the options down step by step:

1. **Start:** 9 theme groups related to the faculty, such as artificial intelligence, cybersecurity, robotics, energy and media.
2. **Do students still want the field?** We dropped media technology and games. In media technology, first-year numbers in Germany changed by -8.6 % a year. We kept robotics, although its related subjects are flat or falling, because it could be combined with artificial intelligence.

8 fields were left.

3. **Is there a gap near OTH?** We dropped general artificial intelligence, because there are already 32 bachelor and 34 master offers within 150 km and OTH offers both. We also dropped themes that OTH already offers, and put microelectronics on a watch list because very few students start it. This left 5 candidates.
4. **Does OTH teach most of it already?** Kill rule 1 removed a master in embedded systems, because OTH already teaches 88 % of its core topics, and a part-time AI master, because OTH already has one. This left 3 options.
5. **Other kill rules:** none of the three failed the rules on staff, falling demand or a falling supply of graduates.
6. **Ranking:** the comparison of the three options in Section 8.

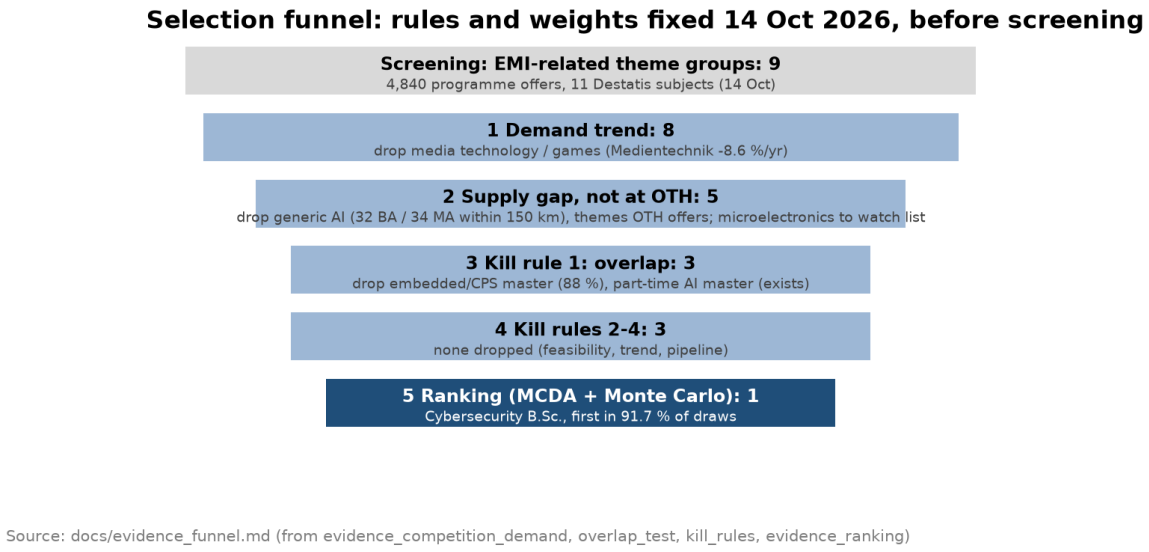


Figure 2: We started with many possible fields and removed them step by step; the weights and kill rules were written in advance.

The funnel is not perfect, and we say where.

- The two screening filters (falling demand, crowded supply) were set on the day of the screening itself.
- The fields that OTH already offers were first dropped without a written reason. A later check gave all 18 themes a written verdict with the same filters. On the same filters no theme came out stronger than cybersecurity; the next step was judged on titles only.
- Robotics stayed in although 50 % of its core topics are already taught in one OTH programme, the AI bachelor. The kill rule only drops options above half.
- The scoring scales for five of the six points, and part of the sixth, were written after we had collected the evidence, but before the first ranking.
- Weak evidence widens a score's range instead of raising it: this rule came before the first ranking; that the widening must be even came after it.

- The bar for calling the result robust was set when we reported the results (Section 8).

6. The four findings

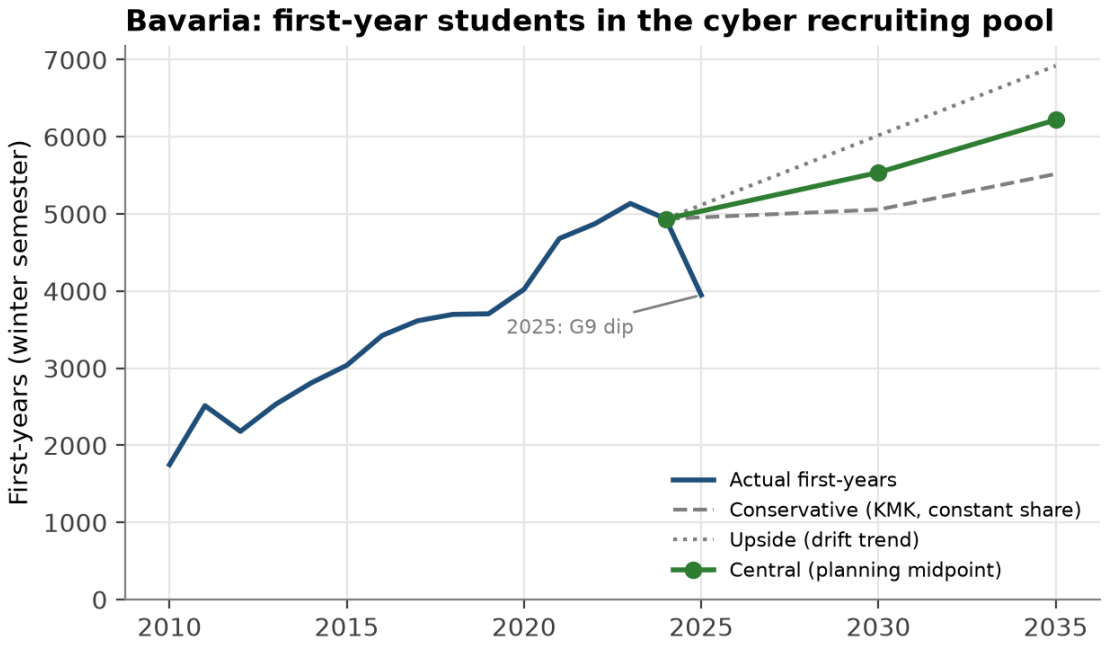
6.1 Do students want it?

What we asked. Is the number of people who want to study this field growing or shrinking?

What we did. Official statistics do not count cybersecurity students separately; they count them under computer science. So we used a **proxy**: we measured the wider field, as you might judge the demand for electric bikes from all bicycle sales. Our proxy is the number of first-year students in computer science and two closely related subjects in Bavaria. To look ahead, we used two ways. The careful way follows the official forecast of first-year students. The trend way continues the growth of the past years. As a planning value we took the middle between the two.

What we found.

- Computer science is the only large subject of the faculty that grew: in Bavaria by 6.1 % a year (2019-2024), with a dip in 2024. In Germany it has been level since 2023.
- Most of that growth is foreign students. In Bavaria, foreign students made up 77 % of the growth. German first-year students grew by only 2.3 % a year in Bavaria, and in Germany they changed by -0.2 %. Among German students, computer science is the only large subject of the faculty that grew in Bavaria.
- Our proxy was 4,936 first-year students in Bavaria in 2024. The careful forecast gives 5,057 for 2030, the middle value 5,538.
- Online, people search for “Cyber Security Studium” more often than for the AI and robotics degree terms (about 2x KI, 2.6x Robotik). But its search interest has hardly grown since 2022.



Source: Destatis 21311-0015; docs/evidence_forecast.md

Figure 3: The number of Bavarian first-year students in computer-related subjects grew strongly and is expected to be back above its 2024 level after the 2025 dip.

How sure are we? Fairly sure for the field as a whole, less sure for cybersecurity itself, because we can only measure it through the wider field. A German-taught bachelor draws mainly German students, and their numbers grow much more slowly.

6.2 Would students come to OTH?

What we asked. How many students would start the programme at OTH each year?

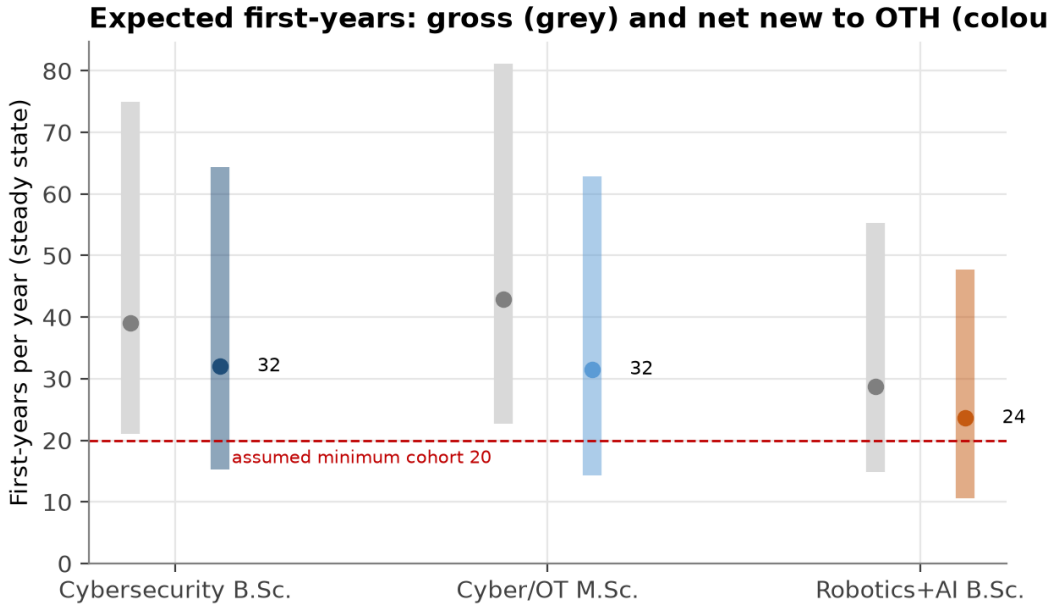
What we did. People tend to shop at the nearest supermarket, and fewer travel far. Students behave in a similar way. We measured how strongly distance puts students off, using real data on which region the students of each state-run Bavarian university of applied sciences come from. With this **capture model** we can say how many students OTH would attract compared with a competitor. But no university of applied sciences we checked publishes how many students start its cybersecurity programme. So we needed **anchors**: known figures of similar programmes that turn the comparison into a number.

A worked example. OTH's **catchment** (the Bavarian counties closer to OTH than to the nearest universities with a cybersecurity programme) has 5,501 school leavers a year who may go to university. About a third of them finish school in Regensburg or its county, where other universities are close by. The model says OTH would attract 0.65-0.86 times the students of TH Ingolstadt. TH Ingolstadt plans its programme for 25 students a year. Depending on the model settings, this gives 16-21

students; we use 21 as our low estimate. Compared with TH Deggendorf, OTH would attract 1.20-1.28 times its students. With the ~60 first-year students from a student review, this gives our high estimate of 75. For the middle estimate we chose a value between the two, 39. This is our choice, not a model result. That is less than one in a hundred of OTH's own school leavers.

What we found.

- 21 / 39 / 75 first-year students a year (low / middle / high) once the programme is fully running; we assume it reaches this level in its third year.
- In the first year: 12 / 24 / 45.
- We assume a programme needs at least 20 first-year students a year; we found no official rule.



Source: data/interim/option_scenarios.csv; docs/evidence_capture.md, docs/evidence_ranking.md

Figure 4: Grey bars show all expected first-year students, coloured bars those who are new to OTH; the cybersecurity bachelor is above the minimum in the middle case.

How sure are we? The comparison between the universities is stable. The weak part is the anchors: a planned capacity and one student review. One real first-year number from TH Ingolstadt or TH Deggendorf would make the range about half as wide.

- Students from outside Bavaria are not in the model, but they matter little. By the same rough method they are about 11.5 % of OTH's German students, 11.0 % at TH Ingolstadt and 8.0 % at TH Deggendorf.
- None of the universities of applied sciences we checked publishes its numbers. Two research universities do (156 and 42 a year), so the ~60 from the student review is a believable high figure. This is a check, not a new anchor.

- TH Deggendorf also runs a regional centre for IT security, so it is a strong competitor in this field.

6.3 Can OTH teach it?

What we asked. How much of a cybersecurity programme does the faculty already teach, and what is missing?

What we did. We compared the **module handbooks** (the official course descriptions) of seven similar programmes with those of the faculty. A computer program read every pair of module descriptions and marked those that teach the same thing. It is like comparing two cookbooks by their dishes, not by their exact words. We checked twenty of its decisions by hand: it was right in 16 of 20. All four of its mistakes counted too much as already taught, so our coverage figures are probably on the high side.

What we found.

- The faculty already teaches up to 71 % of the basics (mathematics, programming, networks). It teaches only about 31 % of the security content (18-39 %, by programme compared).
- Missing are mainly the practical topics: hacking tests, digital forensics, handling security incidents, secure software, security management and law.
- An estimated 2 new professorships, a lab engineer, outside teachers for law and security management, and a separate security lab would close the gap. A draft course plan has 75 (36 %) security credits (**ECTS**) out of 210 ECTS.

OTH's agreement with the state ministry names IT security as one of its research fields.

How sure are we? Fairly sure about the module comparison; we checked a sample of its decisions by hand. The staff figure is an estimate, and we have not estimated the cost of the lab.

6.4 Are there jobs?

What we asked. Would graduates find relevant jobs, in Germany and in the region?

What we did. We counted job adverts on the job board of the Federal Employment Agency, like counting "staff wanted" signs in shop windows. We also looked at official employment statistics and the national report on jobs with too few applicants.

What we found.

- 851 adverts in Germany mentioned "Cyber Security" on one day; 30.9 % of them were in Bavaria, more than the 22-25 % for general IT adverts.
- 191 adverts were within 100 km of Amberg, an area that includes Nuremberg and Regensburg, but only 8 within 50 km.
- IT jobs held by university graduates in Bavaria: +34.3 % from 2019 to 2023. Computer scientists with a bachelor's degree are officially in short supply.

How sure are we? Medium. A job advert is not a job for a graduate, adverts can mention several keywords, and the shortage figure is for computer science in general, not for security alone.

7. Would it just take students from OTH's own programmes?

A new café helps its owner only if it brings in new customers. If it mainly takes customers from the owner's other café down the street, little is gained. A new study programme is the same: students who would have studied at OTH anyway are not new students for OTH. We call this **cannibalisation**.

What we did. We ran the capture model twice, once without and once with the new programme. When a new option opens, the model takes its students from all the other options in proportion to how popular they were before. So we can see where each new student would otherwise have gone: to another OTH programme, or to another university.

A worked example. Of the 39 first-year students in the middle case, about 7 would come from OTH's own programmes with a security focus. The other 32.0 would be new to OTH. About 6.1 of them come from OTH's own region and would otherwise have left it to study elsewhere.

What we found.

- Across the model variants, 14-27 % of the students would come from OTH's own programmes.
- New to OTH: 15.3 / 32.0 / 64.4 students a year (low / middle / high).
- Kill rule 1 is clearly passed: 25 % of the programme's core topics are already taught at OTH; the limit is more than 50 %.

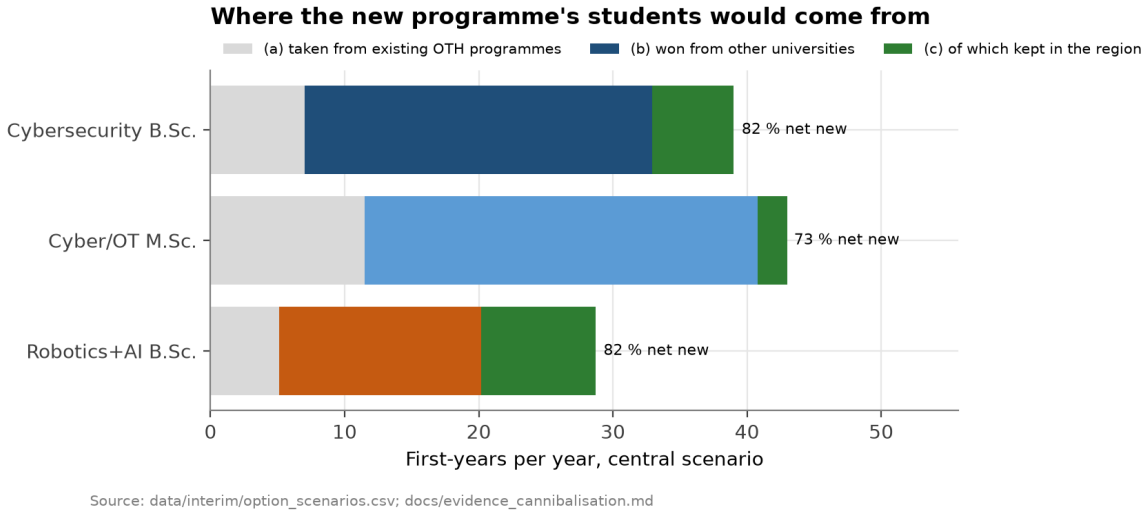


Figure 5: Most of the new programme's students would come from other universities (dark part), not from OTH's own programmes (light part).

How sure are we? Medium. The model only knows OTH programmes that already have a security focus. It does not see a student who would otherwise study artificial intelligence or electrical engineering at OTH. As a stress test we assumed that half of the students come from OTH. Then

10.5 / 19.5 / 37.5 students a year would be new to OTH (low / middle / high). The middle case would then be close to the minimum of 20.

Robotics carries a bigger risk of this kind. 50 % of its core topics are already taught in OTH's artificial intelligence bachelor, on the same campus.

8. Comparing the three options

We compared three finalists: a cybersecurity bachelor, a cybersecurity master with a focus on **OT security** (the security of machines and factories), and a bachelor in robotics with artificial intelligence.

How we scored them. Think of a school report where some subjects count more than others. Each option got a score from 1 to 5 on six points. The points have fixed **weights**: student demand 0.30, labour relevance 0.20, institutional fit 0.20, feasibility 0.15, differentiation 0.10, regional contribution 0.05. Student demand counts most. We recorded no reason when we set the weights, but this fits the task of attracting students. Every score has a range from low to high, and a score that rests on weak evidence got a wider range. With the middle scores, the totals are 3.42 for the cybersecurity bachelor, 3.04 for the master and 2.95 for robotics.

What-if scenarios. Weights and scores are judgements, so a single total could mislead. We therefore repeated the comparison 10,000 times. In each run the weights moved a little around the chosen values, and each score was drawn from its range. Then we counted how often each option came first. It is like replaying a season many times with slightly different weather and seeing which team usually wins.

What we found.

- The cybersecurity bachelor came first in 91.7 % of the scenarios, the master in 6.0 % and robotics in 2.3 %.
- If robotics takes half of its students from OTH (Section 7), the cybersecurity bachelor wins 93.2 % of the scenarios.
- **Sensitivity check:** we left out each scoring point in turn. The cybersecurity bachelor stayed first every time. Its lowest share was 67.2 %, when student demand was left out. So its lead rests mostly on demand, and next on job prospects.
- The master usually comes second (67.0 % of the scenarios); robotics usually comes last (72.2 %).

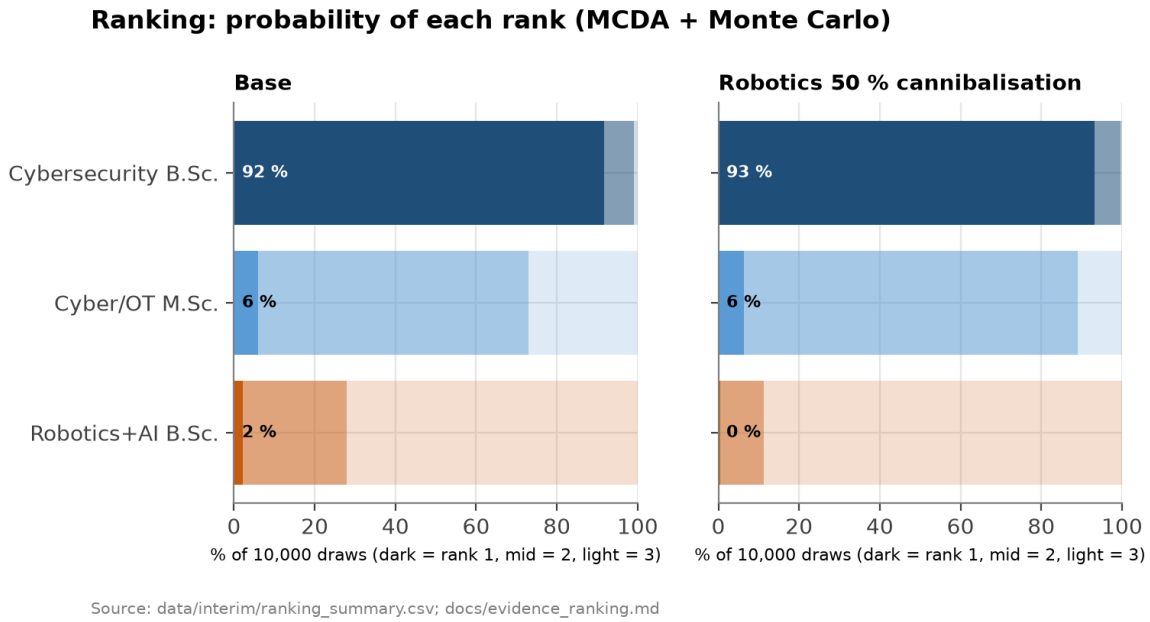


Figure 6: The dark part of each bar shows how often an option came first in the what-if scenarios; cybersecurity wins in most of them.

How sure are we? First place is robust in the base case. If we judge the demand trend on German students only, the demand score of the cybersecurity bachelor drops, and so does robotics'. The bachelor then comes first in 78.2 % of the scenarios, just below our bar for "robust", and the master in 19.5 %. Two more honest notes. In our first run, robotics' wide ranges were uneven and raised its average. Leaving out demand then made robotics and the cybersecurity bachelor about level (49 % / 50 %). We made the ranges even, and the cybersecurity bachelor has stayed first in every check since except one (demand judged on German students and job relevance left out). And the 80 % bar we use for "robust" was set when we reported the results, not before.

9. How sure we are, and the limits

We are fairly sure about the ranking and much less sure about the exact number of students. The limits below are ordered by how much they could change our answer.

1. **No real first-year numbers.** None of the similar programmes we checked publishes how many students start. Our student estimate rests on one planned capacity and one student review. In the low case, only 15.3 of the 21 students a year would be new to OTH. That is below our minimum of 20.
2. **Cybersecurity is measured indirectly.** No official statistic counts cybersecurity students. We read demand from computer science, from online searches and from job adverts.
3. **The capture model is simple.** It uses straight-line distances and only students from Bavaria. Students from outside Bavaria are missing. By the same rough method they are about 11.5 % of OTH's German students, 11.0 % at TH Ingolstadt and 8.0 % at TH Deggendorf, so this

matters little. The model may also understate how many students move between two very similar programmes.

4. **Uneven evidence.** For robotics we could not obtain a verified student number for a comparable programme. We assumed a wide range of first-year numbers for such a programme and gave its scores wider ranges.
5. **Some scoring details came late.** The weights and kill rules came first. The scoring scales for five of the six points, and part of the sixth, were written after we had collected the evidence, but before the first ranking.
6. **Module comparison.** The computer program compares descriptions, not real teaching. It tends to count too much as already taught.
7. **Staff and cost.** The staff need is an estimate, and we did not estimate the cost of the lab. We did not read the research profiles of OTH staff.
8. **Job adverts are signals, not jobs.** One advert can mention several keywords, and the job board does not list every advert.
9. **English-taught teaching.** An English-taught bachelor could reach students from abroad. But the one example we found is weak: the English-taught cyber bachelor at Saarland University had 3 first-year students in winter 2025/26, the German-taught one 42. Our model cannot estimate students from abroad.
10. **Missing sources.** Some databases and pages blocked automatic downloads or needed a login, or our login did not work. We did not work around that. One school year is missing from the school-leaver data.

What we tested. We left out each scoring point in turn. We assumed that robotics takes half its students from OTH. We let the student numbers of the options vary independently, and we gave robotics more favourable ranges. We also judged the demand trend on German students only. The cybersecurity bachelor stayed first. Its lowest share of first places was 67.2 % in the main case and 58.6 % with German students only. Only when we judged demand on German students and also left out job relevance did the master come out narrowly ahead. Not every limit was tested this way. For example, we did not rerun the ranking with half of the cybersecurity students coming from OTH.

Bottom line. The limits mainly affect how big the programme would be, not which option comes first (one combined check puts the master narrowly ahead). That is why our first recommendation is to get one real first-year number before deciding.

10. What we recommend

Our recommendation: OTH should start a bachelor's degree in cybersecurity in the EMI faculty. It was the only finalist we tested in full on all four questions, and it passed each one. It also came first in 91.7 % of the what-if scenarios.

What OTH would need:

- **Students:** plan for about 39 first-year students a year once the programme is fully running (range 21 to 75), and fewer in the first two years.

- **Staff:** 2 new professorships, one for attack and forensics and one for defence and OT security, plus a lab engineer. Outside teachers would cover law and security management.
- **Room:** a separate security lab, ideally linked to the existing automation labs. We have not estimated its cost.
- **Profile:** a broad security bachelor with one module on factory security and a choice of electives. The name should use the English term "Cyber Security", because that is the term people use most when they search for a degree.
- **Start:** after accreditation. Bavaria's switch from eight to nine grammar-school years left one year with few school leavers (2025). That year is already past.

Before deciding: ask TH Ingolstadt and TH Deggendorf, or the Bavarian State Office for Statistics, for the real number of first-year students in their cybersecurity bachelors. If the number is close to TH Ingolstadt's planned 25, OTH should plan carefully and check the low case. If it is close to the figure for TH Deggendorf, our middle estimate is on the careful side.

Two proposals (team judgement, not model results)

(a) The cybersecurity master later. The master came second. We suggest offering it once the bachelor has its own graduates; it could then share its staff and lab, and recruit its graduates. Taught in English, it could also reach foreign students (40 % of computer-science first-years in Germany, 2024). An English-taught bachelor looks weak: at Saarland University it had 3 first-years in winter 2025/26, the German-taught one 42.

(b) Robotics inside the existing AI bachelor. Robotics came last. 50 % of its core topics are already taught in OTH's AI bachelor. If half of its students came from OTH, only 7.5 / 14.3 / 27.6 a year (low / middle / high) would be new to OTH. These numbers rest on an assumed range of student numbers, because we could not obtain a verified figure for a comparable programme. A stronger robotics focus within the AI bachelor would use the strong local robotics job market without a new programme competing with an existing one.

After a start: compare the real number of first-year students, and the share from OTH's region, with our low, middle and high cases every year.

11. Glossary

Anchor: a known figure of a similar programme that turns a comparison into a number of students.

Cannibalisation: the part of a new programme's students who would otherwise have studied in the university's existing programmes.

Capture model: a model of how many students each university attracts, based on how far from it students go to school.

Catchment: the area closer to a university than to its competitors; for OTH, the Bavarian counties closer to it than to the nearest universities with a cybersecurity programme.

ECTS: the European credit points for study work; a full bachelor at OTH has 210 ECTS.

Key numbers: the one generated list from which every statistic in this summary is taken, with the document it comes from.

Kill rule: a rule, written in advance, that removes an option whatever its other strengths.

Module handbook: the official description of every course in a study programme.

OT security: the security of machines, factories and other technical systems, as opposed to office computers.

Professorship: a permanent professor's position, which comes with a fixed amount of teaching.

Proxy: something we can measure that moves together with something we cannot measure directly.

Sensitivity check: a test of whether the answer changes when one assumption is changed or left out.

University of applied sciences: a practice-oriented university such as OTH (Hochschule für angewandte Wissenschaften).

Weights: the importance given to each scoring point; they add up to one.

What-if scenarios: many repeats of the comparison, each with slightly different weights and scores, to see how often each option wins.

12. Where to find the details

Everything here can be traced and rebuilt from github.com/SomanthaManuranga/study-program-intelligence.

What you want	Where it is
The full technical report	built by the pipeline to `reports/final/report.pdf`; its texts are in `docs/report/`
The evidence for each finding	`docs/evidence_*.md`, several documents per question
Method decisions, most with reasons	`docs/decision_log.md`
The rules written in advance	`docs/kill_rules.md` and `configs/analysis.yaml`
Source of each number	`reports/key_numbers.yaml`, built from the evidence documents
All data sources	`configs/sources.yaml`
Answers to likely questions	`docs/defence_qa.md`
An interactive version	the dashboard: `streamlit run app/dashboard.py`
Rebuild everything	`python -m study_program_intelligence.pipeline`, once the files downloaded by hand are in place (README)